Design & Analysis of Algorithms - Dynamic Programming & Gre

Course: Algorithms | Semester 5 | Contributor: Alex Chen

Optimal substructure, memoization, 0/1 Knapsack, Bellman-Ford, and Master Theorem.

Asymptotic Analysis & Recurrences

* Big-O, Omega, and Theta formal definitions with limit evaluation.
* Masters theorem for divide-and-conquer recurrences Tn = aTn/b + fn.
* Substitution method and recursion tree method for complex recurrences.

Dynamic Programming Patterns

* 0/1 Knapsack problem: 2D table state transitions vs fractional greedy approach.
* Longest Common Subsequence LCS and Matrix Chain Multiplication optimal parenthesization.
* Floyd-Warshall all-pairs shortest paths algorithm in OV^3.

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